

Notes: Etula, Rinne, Suominen & Vaittinen (RFS, 2020)
Dash for Cash: Monthly Market Impact of Institutional Liquidity Needs

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1 Big picture

- **Question.** Can predictable recurring payment obligations create systematic price pressure in liquid debt and equity markets around month-end?
- **Mechanism.** Pensions, dividends, salaries, and other transfers cluster near month-end. Institutions sell assets before settlement deadlines to raise cash and later reinvest payments.
- **Key timing.** In U.S. equities and corporate bonds, three-day settlement implies that cash needed on T must be raised by the close of $T-4$; Treasury settlement is shorter, so pressure occurs closer to month-end.
- **Main evidence.** Funding rates rise, stock and bond yields rise, equity returns fall before month-end and rebound afterward, and institutional trading data show net selling before month-end.
- **Economic takeaway.** Synchronized institutional cash needs can move even liquid markets when arbitrage capital is limited.

2 Data and measurement

2.1 Payment-cycle data

Pension payment timing is measured using the 20 largest U.S. public pension plans, and dividend timing uses CRSP dividends. Day T denotes the last trading day of the month.

Detailed interpretation: The payment dates are the primitive cash-flow shocks in the paper.

2.2 Market data

The paper uses repo, Libor, Federal Funds rates, CRSP equity indexes, international equity indexes, Treasury yields, and aggregate bond indexes.

Detailed interpretation: Observing both funding rates and asset prices lets the authors connect the cost of cash to selling pressure in capital markets.

2.3 Institutional and fund data

ANcerno trade data reveal institutional buy-sell imbalances. Mutual fund holdings come from Thomson Reuters, flows from the Investment Company Institute, and hedge fund returns from TASS.

Detailed interpretation: The institutional datasets test whether price pressure is actually associated with institutional liquidity demand.

3 Models and empirical specifications

3.1 Event-time windows

Selling pressure: $T-8$ to $T-4$, Positive reversal: $T-3$ to $T-1$, Reinvestment: T to $T+3$, Negative reversal: $T+4$ to T

Detailed interpretation: Temporary price pressure implies negative correlations between selling-window returns and later reversal-window returns.

3.2 Settlement-rule difference-in-differences

$$Autocorr_{c,after} = \alpha + \beta(Treated_c \times After_c) + \gamma Treated_c + \delta After_c + FE_c + \varepsilon_c.$$

Detailed interpretation: If settlement timing is causal, shortening settlement should shift or weaken the old return autocorrelation.

3.3 Institutional trading regression

$$R_{T-3:T-1} = \alpha + \beta R_{T-8:T-4} + \theta NetSales_{T-8:T-4} + \varepsilon.$$

Detailed interpretation: Positive θ means institutional selling pressure predicts stronger subsequent rebounds.

3.4 Mutual fund flow regression

$$R_{T-8:T-4} = \alpha + \beta Outflow_t + \varepsilon, \quad R_{T-3:T-1} = \alpha + \beta Outflow_t + \gamma R_{T-8:T-4} + \varepsilon.$$

Detailed interpretation: Outflows proxy for aggregate liquidity needs.

4 Identification and empirical design

The paper combines timing predictions from settlement rules, cross-market evidence, international variation, a European settlement reform, direct ANcerno trading evidence, mutual fund holdings and flows, and hedge fund limits-to-arbitrage evidence. Its main limitation is that not every payment obligation or hidden liquidity motive is directly observable.

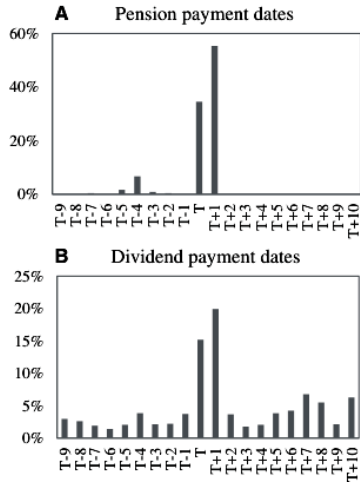
5 Tables and figures

5.1 Figure 1: Payment dates + Figure 2: funding rates

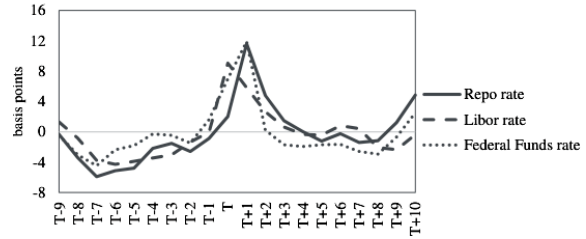
Read: Pension and dividend payments cluster around T and $T + 1$; funding rates rise around the same dates.

Detailed interpretation: These figures establish that the cash-demand shock is real and that the cost of cash responds to it.

Economic message: The month-end payment cycle creates predictable liquidity scarcity.



(a) Figure 1: payment dates



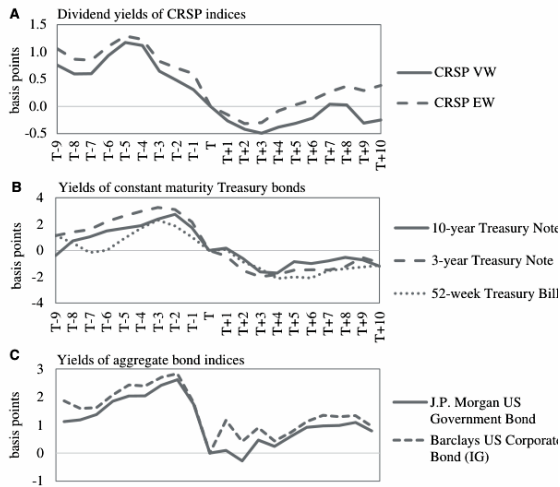
(b) Figure 2: funding rates

5.2 Figure 3: dividend and bond yields + Figure 4: daily returns

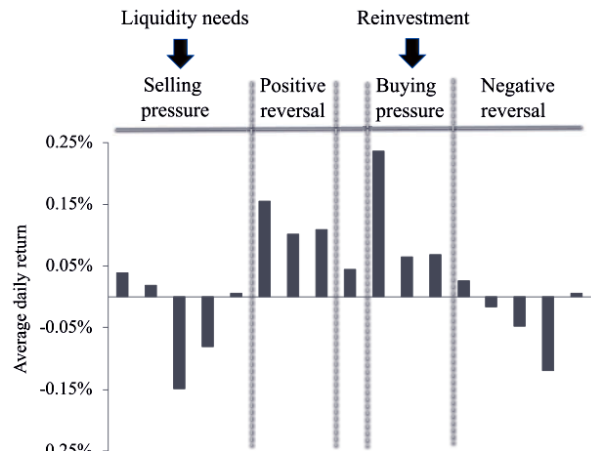
Read: Yields rise before month-end, and equity returns show selling pressure, reversal, reinvestment, and post-reinvestment reversal windows.

Detailed interpretation: The same timing appears in asset prices and returns.

Economic message: The evidence is consistent with temporary price pressure rather than news about fundamentals.



(a) Figure 3: dividend and bond yields



(b) Figure 4: average daily returns

5.3 Figure 5: cumulative stock returns

Read: The $T - 3$ to $T + 3$ window compounds strongly over time, while selling-pressure and other-day windows are much weaker.

Detailed interpretation: Small repeated daily effects become economically important when compounded across decades.

Economic message: The anomaly has large economic magnitude.

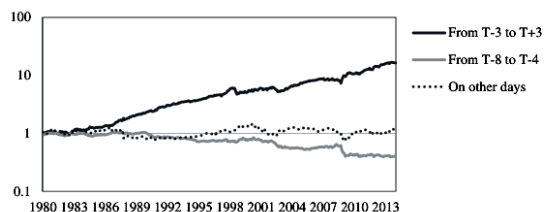


Table 1
Stock market returns near the turn of the month around the world

Country	Settlement period (days)	Sample starts	From T-8 to T-4	From T-3 to T-1	On T	From T+1 to T+3	From T+4 to T+8	Average daily return
United States (S&P 500)	3	Jul. 1985	-0.04%	0.11%	-0.00%	0.13%	-0.03%	0.04%
United States (CRSP VW)	3	Jul. 1995	-0.03%	0.12%	0.04%	0.12%	-0.03%	0.04%
United States (S&P 500)	5 or 3	Jan. 1980	-0.01%	0.11%	0.00%	0.13%	-0.01%	0.05%
United States (CRSP VW)	5 or 3	Jan. 1980	-0.02%	0.11%	0.14%	0.12%	-0.01%	0.05%
Other developed countries								
Australia (S&P/ASX200)	3	Feb. 1999	-0.02%	0.14%	0.10%	0.09%	-0.02%	0.04%
Austria (ATX)	3	Feb. 1998	0.00%	0.18%	0.17%	0.19%	-0.07%	0.05%
Belgium (BEL20)	3	Jan. 1990	-0.05%	0.00%	0.20%	0.15%	-0.03%	0.03%
Canada (S&P/TSX C)	3	July 1995	0.00%	0.00%	0.21%	0.09%	-0.03%	0.04%
Denmark (OMXC20)	3	Dec. 1989	-0.06%	0.07%	0.14%	0.18%	0.00%	0.04%
Finland (OMXHEX25)	3	Jan. 1991	-0.02%	0.12%	0.33%	0.16%	-0.02%	0.06%
France (CAC40)	3	Oct. 2000	-0.07%	0.18%	0.15%	0.07%	-0.11%	0.01%
Iceland (ISEQ OVER)	3	Mar. 2001	-0.05%	0.02%	0.35%	0.17%	-0.05%	0.01%
Italy (FTSE MIB)	3	Jan. 1998	-0.06%	0.12%	0.09%	0.08%	-0.07%	0.02%
Japan (NIKKEI225)	3	Jan. 1980	0.01%	0.11%	0.11%	0.07%	-0.06%	0.02%
Netherlands (AEX)	3	Jan. 1983	0.00%	0.07%	0.15%	0.18%	-0.02%	0.05%
New Zealand (NZX50)	3	Jan. 2001	0.00%	0.12%	0.25%	0.07%	-0.04%	0.03%
Norway (OSL)	3	Jan. 1987	-0.02%	0.07%	0.26%	0.17%	-0.02%	0.05%
Portugal (PSI-20)	3	Dec. 1998	-0.08%	0.05%	0.14%	0.12%	-0.01%	0.00%
Singapore (STI)	3	Sep. 1999	-0.02%	0.10%	0.17%	0.18%	0.01%	0.03%
Spain (IBEX35)	3	Mar. 1997	-0.06%	0.11%	0.10%	0.18%	-0.07%	0.04%
Sweden (OMXS30)	3	Jan. 1986	-0.03%	0.12%	0.16%	0.28%	0.00%	0.06%
Switzerland (SMI)	3	Jul. 1988	-0.04%	0.09%	0.11%	0.15%	-0.01%	0.04%
the United Kingdom (FTSE100)	3	Aug. 1996	-0.04%	0.11%	0.04%	0.16%	-0.03%	0.03%
Countries with a settlement period less than 3 days								
Germany (DAX)	2	Jan. 1980	-0.03%	0.06%	0.16%	0.21%	-0.04%	0.05%
Hong Kong (HSI)	1 or 2	Jan. 1980	-0.03%	0.08%	0.26%	0.15%	0.03%	0.07%
Israel (TA-25)	1	Jan. 1992	0.00%	0.08%	0.17%	0.17%	0.04%	0.06%
Average of all indices excluding the United States			-0.03%	0.10%	0.17%	0.14%	-0.03%	0.04%

This table presents average daily stock market returns near the turn of the month in the United States and in other developed countries as defined by FTSE, MSCI, and S&P. T refers to the last trading day of the month. Our sample starts in January 1980 or later as the relevant data become available, and settlement rule is 3 days or less. For the United States, we show also the full sample results. The sample runs until the end of 2013. All figures statistically significant at the 5% level are displayed in bold.

5.4 Tables 1–2: international returns and correlations

Read: Many developed markets display positive reversal-window returns and negative selling-to-reversal correlations.

Detailed interpretation: The pattern is not only a U.S. phenomenon.

Economic message: The month-end liquidity cycle appears global.

Table 2
Return correlations near the turn of the month around the world

Country	Settlement period (days)	Sample starts	Correlation of T-8 to T-4 and T-3 to T-1 returns	Correlation of T to T+3 and T+4 to T+8 returns	Daily return autocorrelation	Weekly return autocorrelation
United States (S&P 500)	3	Jul. 1995	-0.38	-0.11	-0.07	-0.08
United States (CRSP VW)	3	Jul. 1995	-0.39	-0.06	-0.04	-0.06
United States (S&P 500)	5 or 3	Jan. 1980	-0.30	-0.09	-0.03	-0.05
United States (CRSP VW)	5 or 3	Jan. 1980	-0.32	-0.03	0.01	-0.02
Other developed countries						
Australia (S&P/ASX200)	3	Feb. 1999	-0.35	-0.16	-0.04	-0.06
Austria (ATX)	3	Feb. 1998	-0.41	-0.09	0.06	-0.01
Belgium (BEL20)	3	Jan. 1990	-0.26	-0.23	0.07	-0.01
Canada (S&P/TSX C)	3	Jul. 1995	-0.27	0.03	0.00	-0.09
Denmark (OMXC20)	3	Dec. 1989	-0.38	-0.02	0.06	-0.05
Finland (OMXHEX25)	3	Jan. 1991	-0.01	-0.19	0.04	0.02
France (CAC40)	3	Oct. 2000	-0.42	-0.18	-0.04	-0.09
Iceland (ISEQ OVER)	3	Mar. 2001	-0.26	-0.32	0.05	-0.05
Italy (FTSE MIB)	3	Jan. 1998	-0.28	-0.04	0.00	-0.01
Japan (NIKKEI225)	3	Jan. 1980	-0.18	0.00	-0.02	-0.02
Netherlands (AEX)	3	Jan. 1983	-0.21	0.00	0.03	0.03
New Zealand (NZX50)	3	Jan. 2001	-0.03	0.05	0.05	0.04
Norway (OSL)	3	Jan. 1987	-0.23	-0.10	0.03	0.02
Portugal (PSI-20)	3	Dec. 1998	-0.21	-0.16	0.08	0.01
Singapore (STI)	3	Sep. 1999	-0.25	-0.06	0.03	0.03
Spain (IBEX35)	3	Mar. 1997	-0.21	-0.15	0.02	-0.06
Sweden (OMXS30)	3	Jan. 1986	-0.11	-0.11	0.04	-0.02
Switzerland (SMI)	3	Jul. 1988	-0.17	-0.24	0.03	-0.07
the United Kingdom (FTSE100)	3	Aug. 1996	-0.33	-0.24	-0.03	-0.08
Countries with a settlement period less than 3 days						
Germany (DAX)	2	Jan. 1980	-0.15	-0.15	0.00	-0.02
Hong Kong (HSI)	1 or 2	Jan. 1980	-0.19	-0.04	0.03	0.08
Israel (TA-25)	1	Jan. 1992	-0.13	-0.08	0.02	-0.07
Average of all indices excluding the United States			-0.24	-0.12	0.02	-0.02

This table presents correlations of returns between $T-8$ to $T-4$ and $T-3$ to $T-1$, and correlations of returns between T to $T+3$ and $T+4$ to $T+8$. T refers to the last trading day of the month. Our sample starts in January 1980 or later as the relevant data become available, and settlement rule is 3 days or less. For the United States, we also show the full sample results. The sample runs until the end of 2013. All figures statistically significant at the 5% level are displayed in bold.

Table 2: return correlations around the world

5.5 Tables 3–4: settlement and Treasury evidence

Read: The European settlement reform weakens old timing correlations, and Treasury yields exhibit reversal behavior.

Detailed interpretation: Settlement conventions determine when liquidity demand hits prices.

Economic message: Market plumbing matters for asset pricing.

Table 3
Difference-in-differences test around a change in the settlement period

Autocorrelation at $T-2$		
Treatment group * After change	-0.777 (-3.66)	-0.777 (-3.20)
Treatment group	0.178 (1.31)	
After change	0.336 (1.80)	0.336 (1.51)
Intercept	-0.313 (-2.84)	
Country fixed effects	No	Yes
N	40	40
R ²	.397	.786

This table shows the results from a difference-in-differences test whether a change in the settlement period affects market index return autocorrelations at $T-2$ (i.e., correlation of $T-3$ and $T-2$ market returns). In October 6 2014, most of the European countries changed their settlement period from 3 to 2 days. Our treatment group is formed from our international sample countries affected by this change (AUT, BEL, CHE, DNK, FIN, FRA, GBR, IRL, ITA, NLD, NOR, PRT, and SWE). Our control group consists of all countries in our international sample following the 3-day settlement period at the end of September 2013 and not affected by this change (AUS, CAN, ESP, JPN, NZL, SGP, and USA). In the first regression, autocorrelation at $T-2$ is regressed on treatment group dummy, after change dummy and their interaction, and in the second specification autocorrelation is regressed on after change dummy, treatment dummy (Treatment group dummy * after change dummy), and country fixed effects. T refers to the last trading date of the month. Autocorrelations are calculated using 1 year of data before and after October 2014. t -statistics based on White heteroscedasticity-robust standard errors are shown below the coefficients. All figures statistically significant at the 5% level are displayed in bold.

(a) Table 3: settlement DiD

Table 4
Return reversals before the turn of the month: Evidence from Treasury yields

$y =$ yield change from $T-3$ to $T-1$	52-week Treasury Bill	3-year Treasury Note
$T-8$ to $T-4$ yield change	-0.189 (-2.48)	-0.186 (-2.58)
Intercept	-0.008 (-1.00)	-0.005 (-0.69)
R ²	.081	.043

This table shows the results from a regression in which the changes in Treasury yields from $T-3$ to $T-1$ are regressed on the changes in Treasury yields from $T-8$ to $T-4$. Here, T refers to the last trading day of the month. The Treasury yields are those of 52-week Treasury bill and 3-year Treasury note. Both regressions include thirteen unreported dummy variables that occur in the same maturity during days $T-9$ to $T+3$. The dummy variables control for the Treasury auction effects documented in Lou, Yan, and Zhang (2013): significant price pressure during 4 days preceding Treasury auctions and the subsequent 6-day return reversal. The yield data come from Datastream. The sample period ranges from July 1995 to December 2013. t -statistics based on Newey and West (1987) standard errors are shown below the coefficients. All figures statistically significant at

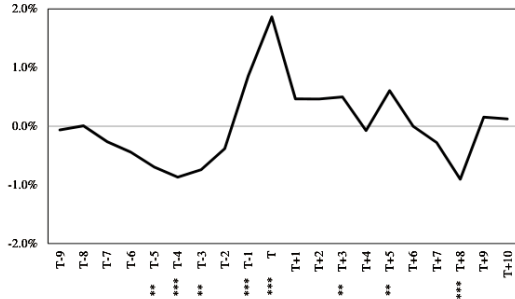
(b) Table 4: Treasury yield reversals

5.6 Figures 6–7 and Table 5: direct institutional trading evidence

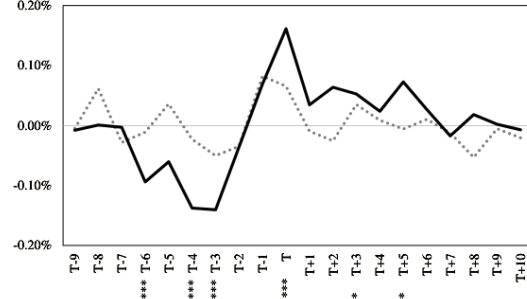
Read: Institutions sell before month-end, identified liquidity demanders show persistent selling, and institutional net sales predict stronger rebounds.

Detailed interpretation: This is the paper's most direct evidence linking institutional trading to market-wide price pressure.

Economic message: Institutional liquidity demand moves prices.



(a) Figure 6: buy ratios



(b) Figure 7: institutional trading patterns

Table 5
Impact of institutional trading on the turn of the month returns

y = market return from $T-3$ to $T-1$				
Market return ($T-8$ to $T-4$)	-0.352 (-2.51)		-0.344 (-2.71)	-0.344 (-2.71)
Institutional investors' net sales ($T-8$ to $T-4$)	35.27 (1.50)		40.37 (2.14)	
Institutional investors' net sales ($T-5$ to $T-4$)		117.25 (2.53)		111.09 (3.49)
Intercept	0.003 (2.39)	0.002 (0.98)	0.000 (-0.09)	0.001 (0.32)
R^2	.184	.020	.084	.209

This table shows the results from a regression in which the market index return from $T-3$ to $T-1$ is regressed on the $T-8$ to $T-4$ index return, and on institutional investors' net sales during $T-8$ to $T-4$. Here, T refers to the last trading day of the month. Institutional investors' net sales is defined as the difference between the value of all ANcerno institutions' stock sales and purchases during the days $T-8$ to $T-4$, or, alternatively, during the days $T-5$ to $T-4$, when the difference is positive, and 0 otherwise. The figures are normalized by the U.S. total stock market capitalization at the beginning of the selling pressure period. Our institutional investors' trade data come from ANcerno, and the sample period ranges from January 1999 to December 2013. The index returns are those of the CRSP value-weighted index. t -statistics based on Newey and West (1987) standard errors are shown below the coefficients. All figures statistically significant at the 5% level are displayed in bold.

Table 5: institutional trading and returns

5.7 Table 6 and Figures 8–10: mutual fund ownership and stock characteristics

Read: Cross-sectional reversals are stronger among stocks held by mutual funds and among large/liquid stocks.

Detailed interpretation: Mutual funds transmit payment-cycle liquidity pressure, and institutions choose low-cost-to-trade stocks to raise cash.

Economic message: Delegated portfolio management creates price pressure and hidden trading costs.

Table 6
Cross-sectional return reversals around the turn of the month

A. Deciles based on returns from $T-8$ to $T-4$											
	1	2	3	4	5	6	7	8	9	10	1-10
Returns from $T-3$ to $T-1$	0.98% (3.76)	0.72% (3.50)	0.60% (3.31)	0.58% (3.49)	0.53% (3.45)	0.48% (3.23)	0.46% (3.12)	0.44% (2.98)	0.41% (2.67)	0.26% (1.44)	0.72% (4.19)
Returns from T to $T+3$	0.65% (2.16)	0.52% (2.29)	0.52% (2.70)	0.46% (2.65)	0.39% (2.34)	0.42% (2.53)	0.43% (2.57)	0.36% (2.09)	0.41% (2.11)	0.31% (1.28)	0.34% (1.78)

B. Deciles based on returns from T to $T+3$											
	1	2	3	4	5	6	7	8	9	10	1-10
Returns from $T+4$ to $T+8$	0.13% (0.43)	-0.10% (-0.41)	-0.12% (-0.57)	-0.15% (-0.73)	-0.10% (-0.55)	-0.15% (-0.83)	-0.14% (-0.73)	-0.16% (-0.83)	-0.19% (-0.92)	-0.33% (-1.29)	0.46% (2.66)

Panel A shows evidence of cross-sectional return reversals around the turn of the month by displaying the returns from $T-3$ to $T-1$ and from T to $T+3$ for deciles of stocks based on their $T-8$ to $T-4$ returns. Here, T refers to the last trading day of the month. In Panel B, the table shows returns from $T+4$ to $T+8$ for deciles of stocks based on their T to $T+3$ returns. Our sample includes all stocks in CRSP that have a share price above 5 USD, and a market capitalization that exceeds NYSE tenth market capitalization percentile on the tenth trading day of the corresponding month. The sample period ranges from July 1995 to December 2013. The last column shows the difference in returns between the two extreme deciles. t -statistics are shown in the parentheses. All figures

Table 6: cross-sectional return reversals

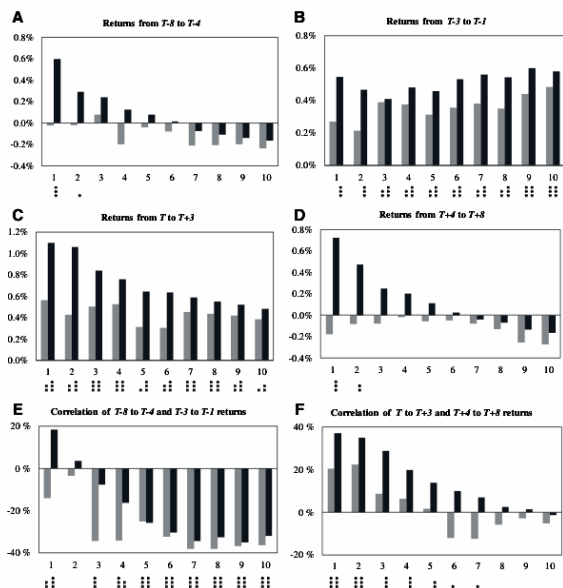
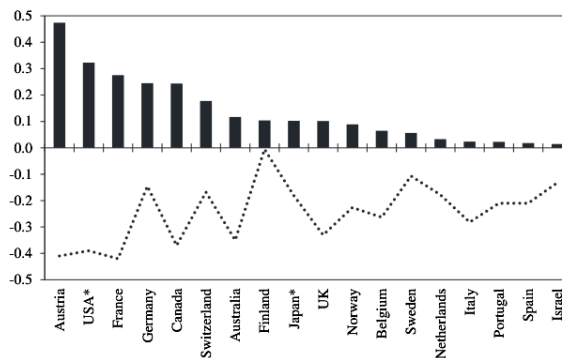


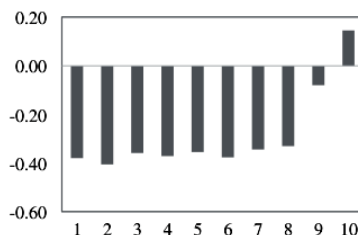
Figure 8

(a) Figure 8: mutual fund holdings



(b) Figure 9: mutual fund ownership across countries

A Stock-level liquidity and T-8 to T-4 and T-3 to T-1 return cor



B Size and T-8 to T-4 and T-3 to T-1 return correlation

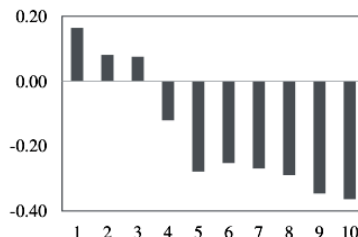


Figure 10: liquidity, size, and correlations

5.8 Tables 7–10: mutual fund AUM, flows, alphas, and characteristics

Read: Mutual fund AUM and outflows strengthen reversals; funds more exposed to the reversal pattern earn lower alphas; high-cash funds are less exposed.

Detailed interpretation: These results connect the market anomaly to mutual fund liquidity management and fund performance.

Economic message: Month-end liquidity management affects investor welfare.

Table 7
Impact of mutual funds' AUM on month-end return reversals

y = market returns from $T-3$ to $T-1$			
Market return ($T-8$ to $T-4$)	-0.315 (-2.58)	0.627 (1.52)	0.627 (1.53)
Mutual fund industry AUM		0.043 (1.51)	0.048 (0.32)
Interaction of mutual fund industry AUM and market return ($T-8$ to $T-4$)		-4.450 (-1.96)	-4.446 (-1.98)
Linear trend			0.006 (-0.03)
Intercept	0.003 (2.55)	0.006 (-0.98)	0.006 (-0.31)
R^2	.150	.200	.200

This table shows the results from a regression of market index returns from $T-3$ to $T-1$ on $T-8$ to $T-4$ index returns, and on mutual fund industry AUM, and its interaction with the $T-8$ to $T-4$ index returns. T refers to the last trading day of the month. Mutual fund industry AUM is the sum of all domestic equity mutual funds' assets under management based on the CRSP mutual fund database, normalized by the U.S. total stock market capitalization. The index returns are those for the CRSP value-weighted index. The sample period ranges from July 1995 to December 2013. t -statistics based on Newey-West standard errors are shown below the coefficients. All figures statistically significant at the 5% level are displayed in bold.

(a) Table 7: AUM and reversals

Table 9
Mutual fund alphas and exposures to month-end return reversals

Mutual fund deciles based on correlation of $T-8$ to $T-4$ and $T-3$ to $T-1$ returns										
	1	2	3	4	5	6	7	8	9	10-1
CAPM										
α	-2.42% (-1.91)	-2.86% (-2.93)	-2.50% (-3.22)	-1.78% (-2.58)	-0.11% (-0.17)	0.67% (0.89)	1.50% (1.47)	3.17% (2.31)	4.04% (2.44)	4.84% (2.24)
$R_M - R_F$	1.07 (28.99)	1.13 (39.60)	1.10 (50.33)	1.09 (62.63)	1.06 (72.41)	1.02 (54.79)	1.01 (39.73)	0.99 (25.98)	0.95 (20.92)	0.84 (16.06)
Fama-French 3-factor model										
α	-2.48% (-2.16)	-2.65% (-3.43)	-2.56% (-3.84)	-2.10% (-3.44)	-0.90% (-1.77)	-0.51% (-0.91)	-0.11% (-0.16)	0.94% (0.96)	1.39% (1.18)	2.37% (1.41)
$R_M - R_F$	1.04 (28.24)	1.10 (52.87)	1.07 (65.45)	1.06 (79.97)	1.02 (85.45)	0.99 (69.12)	0.96 (50.45)	0.93 (35.54)	0.89 (27.10)	0.78 (16.85)
SMB	0.13 (2.59)	0.12 (3.78)	0.14 (3.71)	0.16 (5.46)	0.19 (8.84)	0.20 (8.68)	0.25 (9.87)	0.29 (8.56)	0.33 (8.88)	0.34 (5.79)
HML	-0.09 (-1.66)	-0.13 (-3.71)	-0.09 (-2.96)	-0.06 (-2.11)	-0.06 (0.35)	0.01 (2.03)	0.07 (2.57)	0.11 (3.36)	0.20 (3.40)	0.29 (2.26)
Fama-French 4-factor model										
α	-2.51% (-2.13)	-2.63% (-3.37)	-2.56% (-3.79)	-2.13% (-3.46)	-0.91% (-1.78)	-0.58% (-1.02)	-0.16% (-0.22)	0.88% (0.87)	1.28% (1.05)	2.15% (1.24)
$R_M - R_F$	1.05 (26.74)	1.10 (46.11)	1.07 (51.41)	1.06 (62.28)	1.03 (66.89)	1.00 (54.65)	0.97 (43.32)	0.94 (30.11)	0.91 (23.68)	0.83 (15.86)
SMB	0.13 (2.60)	0.12 (3.74)	0.14 (3.67)	0.16 (5.44)	0.19 (8.78)	0.20 (9.13)	0.25 (10.11)	0.29 (8.81)	0.33 (9.55)	0.34 (6.17)
HML	-0.09 (-1.65)	-0.13 (-3.71)	-0.09 (-3.01)	-0.06 (-2.19)	-0.06 (0.34)	0.01 (2.03)	0.07 (2.56)	0.11 (3.32)	0.24 (3.38)	0.19 (2.27)
MOM	0.01 (0.26)	0.01 (-0.50)	0.01 (-0.03)	0.01 (0.62)	0.01 (0.33)	0.03 (1.49)	0.02 (1.03)	0.02 (0.81)	0.04 (1.42)	0.09 (2.97)

This table shows active domestic equity mutual funds' annualized alphas conditional on fund-specific trailing two-year correlations between the funds' $T-8$ to $T-4$ and $T-3$ to $T-1$ returns. More specifically, funds are divided into deciles every month based on this correlation. Alphas are calculated using the monthly returns of equal-weighted mutual fund portfolios controlling for standard risk factors with

(a) Table 9: alphas and exposures

Table 8
Impact of mutual fund outflows on the turn of the month returns

A. Impact of outflow on $T-8$ to $T-4$ returns		
y = returns from $T-8$ to $T-4$		
Mutual fund industry outflow	-190.00 (-2.87)	-176.34 (-2.55)
Past 20-day returns		0.052 (0.53)
Intercept	0.001 (0.28)	0.000 (0.10)
R^2	.189	.193
B. Impact of outflow on $T-3$ to $T-1$ returns		
y = returns from $T-3$ to $T-1$		
Mutual funds industry outflow	212.51 (4.30)	136.43 (2.63)
$T-8$ to $T-4$ return		-0.345 (-3.00)
Intercept	0.002 (0.81)	0.002 (1.09)
R^2	.296	.437

This table shows the results from a regression in which the market index returns from $T-8$ to $T-4$ (panel A) or $T-3$ to $T-1$ (panel B) are regressed on the past market index returns, and on the mutual fund industry outflow. Here, T refers to the last trading day of the month. Mutual fund industry outflow (normalized by the U.S. total stock market capitalization) is defined as the negative of the aggregate net flow to mutual fund industry (equity, hybrid, and bond funds; see footnote 38 for motivation) from the first Wednesday of the month until the last Wednesday before $T-8$ (panel A), or until the last Wednesday before $T-3$ (panel B), when the net flow is negative, and 0 otherwise. Our weekly mutual funds' flow data come from Investment Company Institute, and the sample period ranges from January 2007 to December 2013. The index returns are those of the CRSP value-

(b) Table 8: outflows and returns

Table 10
Mutual fund characteristics and exposures to month-end return reversals

Mutual fund deciles based on correlation of $T-8$ to $T-4$ and $T-3$ to $T-1$ returns										
	1	2	3	4	5	6	7	8	9	10-1
Trailing 2-year correlation of $T-8$ to $T-4$ and $T-3$ to $T-1$ returns										
Correlation	-0.45	-0.37	-0.34	-0.31	-0.29	-0.26	-0.23	-0.19	-0.14	-0.02
Annualized mutual fund return in excess of the risk-free rate										
$T-8$ through $T-4$	-4.51%	-5.25%	-5.01%	-4.79%	-4.44%	-4.04%	-3.87%	-3.32%	-3.09%	2.52%
$T-3$ through $T-1$	5.29%	5.25%	5.30%	5.62%	6.16%	6.13%	6.63%	7.06%	7.13%	1.61%
Other days	0.87%	1.39%	1.43%	1.58%	2.30%	2.61%	2.72%	3.28%	3.76%	2.45%
All days	1.35%	1.10%	1.38%	2.07%	3.65%	4.30%	5.07%	6.61%	7.32%	6.37%
Mutual fund portfolio composition during the ranking month										
Cash-%	3.47%	3.03%	3.26%	3.28%	3.28%	3.19%	3.24%	3.53%	4.02%	6.57%
Equity-%	91.77%	93.03%	93.11%	92.79%	93.42%	93.15%	93.26%	92.58%	91.77%	87.60%
Bond-%	0.62%	0.57%	0.54%	0.55%	0.57%	0.58%	0.65%	0.67%	0.66%	0.76%
Other mutual fund characteristics										
Size	1,346	1,625	1,421	1,316	1,241	1,272	1,146	1,156	1,129	884
Median size	251	293	275	259	252	242	239	232	212	193
Active share	78.6%	74.7%	73.9%	73.9%	74.9%	76.0%	79.0%	81.2%	83.5%	87.9%
Institutional fund	20.5%	25.3%	26.8%	27.7%	27.5%	27.2%	26.1%	24.4%	22.8%	18.3%
Turnover	113.7%	94.3%	92.0%	91.3%	93.3%	93.8%	93.4%	94.6%	98.2%	134.0%
Expense ratio	1.37%	1.26%	1.24%	1.24%	1.26%	1.26%	1.29%	1.31%	1.35%	1.51%

This table shows the active equity mutual funds' returns and characteristics conditional on fund-specific trailing

(b) Table 10: fund characteristics

5.9 Table 11 and Figure 11: hedge funds and funding constraints

Read: Hedge funds do not fully eliminate the price pressure; some strategies provide liquidity, but high TED-spread months show stronger reversals.

Detailed interpretation: Arbitrage is state-dependent and constrained by funding liquidity.

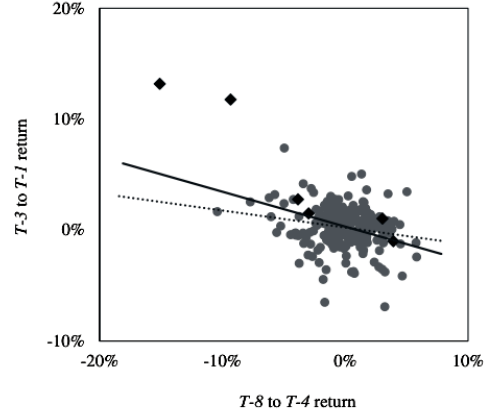
Economic message: Predictable liquidity shocks persist because arbitrage capital is limited.

Table 11
Hedge funds' liquidity provision around the turn of the month

	All funds	During high TED spread	During low TED spread	Global Macro	Managed Futures
$T-5$	-0.114 (-16.15)	-0.078 (-8.90)	-0.044 (-3.73)	-0.031 (-0.75)	0.145 (4.12)
$T-4$	-0.091 (-12.71)	-0.111 (-11.93)	-0.216 (-19.20)	-0.084 (-1.76)	0.082 (2.19)
$T-3$	-0.025 (-3.75)	-0.047 (-5.66)	0.057 (4.29)	0.094 (2.05)	0.393 (9.63)
$T-2$	-0.087 (-13.28)	-0.142 (-15.16)	-0.004 (-0.42)	-0.065 (-1.88)	-0.041 (-1.10)
$T-1$	-0.062 (-10.42)	-0.056 (-8.81)	-0.071 (-7.28)	0.082 (2.12)	-0.048 (-1.08)
T	-0.177 (-21.29)	-0.244 (-24.89)	0.054 (4.10)	-0.120 (-2.47)	-0.128 (-2.46)
$T+1$	0.136 (20.69)	0.249 (30.66)	0.048 (5.13)	0.037 (0.96)	0.181 (4.58)
$T+2$	0.259 (33.88)	0.381 (37.35)	0.176 (18.06)	0.073 (1.57)	0.249 (6.39)
$T+3$	0.168 (25.01)	0.231 (24.17)	-0.010 (-1.09)	0.112 (2.47)	0.109 (2.43)
$T+4$	0.103 (16.30)	0.069 (8.17)	0.107 (12.25)	0.012 (0.29)	-0.078 (-2.48)
$T+5$	0.043 (5.73)	0.034 (3.99)	0.163 (13.95)	0.006 (0.17)	-0.038 (-0.93)
N	7,732	5,183	3,853	312	515

This table shows the hedge funds' average excess market betas around the turn of the month in selected hedge fund style categories and during low (below sample median) and high TED spread. T refers to the last trading day of the month. Hedge funds' average excess market betas are based on fund-specific regressions in which hedge fund's (monthly) return is regressed on the daily S&P 500 index returns around the turn of the month ($T-5$ to $T+5$) and the return on the S&P 500 index outside the turn of the month period. Excess market betas for any given fund are calculated as the difference of its estimated beta for any given day and its beta outside the turn of the month period. Hedge fund data come from TASS, and our sample period is from July 1995 to December 2013. t -statistics are shown below the coefficients. All figures statistically significant at the 5% level are displayed in bold.

(a) Table 11: hedge fund liquidity provision



(b) Figure 11: funding constraints

6 Result map

- Payment dates create synchronized cash demand.
- Settlement rules map payment dates into forced-selling dates.
- Funding rates and asset yields show the price of cash and asset prices respond.
- Direct institutional trades show selling before month-end and buying after.
- Mutual fund holdings and flows show where the effect is strongest.
- Hedge fund evidence explains why arbitrage is incomplete.

7 Short summary

The paper shows that recurring payment obligations create predictable liquidity shocks that temporarily move funding rates, equity prices, and bond prices. Institutions sell before month-end to raise cash, markets rebound as pressure abates, and mutual funds and hedge-fund constraints help explain both the mechanism and the persistence of the anomaly.

8 Conclusion

8.1 Core conclusion

The monthly payment cycle generates predictable and economically important price pressure in liquid markets.

8.2 Economic intuition

Synchronized payment obligations force institutions to sell assets before settlement deadlines, depressing prices. Once the cash need is met and reinvestment arrives, prices rebound.

8.3 Why the evidence is persuasive

The paper combines payment-date evidence, funding-rate evidence, yield evidence, return patterns, settlement variation, institutional trades, mutual fund holdings and flows, and hedge fund funding constraints.

8.4 Implications

Institutions could reduce costs by changing liquidity management; regulators can reduce pressure through shorter settlement; central banks may monitor month-end funding stress; investors can understand turn-of-month returns as compensation for absorbing institutional liquidity shocks.

8.5 Limitations

Not every cash need is observable, ANcerno is incomplete, and the trading-cost estimates rely on counterfactual timing assumptions.

8.6 Takeaway

Calendar-driven cash-flow frictions and limited arbitrage can generate systematic price pressure even in highly liquid markets.